

## SAFETY DATA SHEET

Revision Date 01-April-2024

Revision Number 3

### 1. Identification

**Product Name** 4-Chlorobenzylmagnesium chloride, 0.50M in 2-Me-THF

**Cat No. :** H54625

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |              |
|---------------------------------------------------------|--------------|
| <b>Flammable liquids</b>                                | Category 2   |
| <b>Acute oral toxicity</b>                              | Category 4   |
| <b>Skin Corrosion/Irritation</b>                        | Category 1 B |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 1   |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3   |
| Target Organs - Respiratory system.                     |              |
| <b>Physical Hazards Not Otherwise Classified</b>        | Category 1   |
| May form explosive peroxides                            |              |

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**

Highly flammable liquid and vapor  
Harmful if swallowed  
Causes severe skin burns and eye damage  
May cause respiratory irritation  
May form explosive peroxides



#### Precautionary Statements

##### Prevention

Keep container tightly closed  
Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Ground/bond container and receiving equipment  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection  
Use non-sparking tools  
Take action to prevent static discharges

##### Response

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Rinse mouth  
Do NOT induce vomiting  
Wash contaminated clothing before reuse  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

##### Storage

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

##### Disposal

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                        | CAS-No   | Weight % |
|----------------------------------|----------|----------|
| 2-Methyltetrahydrofuran          | 96-47-9  | 89.17    |
| 4-Chlorobenzylmagnesium chloride | 874-72-6 | 10.83    |

### 4. First-aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.                                                                                                                                                                                                                                                       |
| <b>Inhalation</b>                      | If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.                                                                                                                |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.                                                                                                                                                                                                                                                                                                          |
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                                                                         |

## 5. Fire-fighting measures

|                                         |                                                                                                                                                                                                                                                                                          |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Carbon dioxide (CO <sub>2</sub> ). Powder. Water spray. In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion. CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                                                                                                                                                                 |
| <b>Flash Point</b>                      | No information available                                                                                                                                                                                                                                                                 |
| <b>Method -</b>                         | No information available                                                                                                                                                                                                                                                                 |
| <b>Autoignition Temperature</b>         | No information available                                                                                                                                                                                                                                                                 |
| <b>Explosion Limits</b>                 |                                                                                                                                                                                                                                                                                          |
| <b>Upper</b>                            | No data available                                                                                                                                                                                                                                                                        |
| <b>Lower</b>                            | No data available                                                                                                                                                                                                                                                                        |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                                                                                                                                                                 |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                                                                                                                                                 |

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen chloride. Magnesium oxides.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 3             | 3                   | 2                  | W                       |

## 6. Accidental release measures

|                                  |                                                                                                                                                                                                                                                      |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges. |
| <b>Environmental Precautions</b> | Should not be released into the environment. See Section 12 for additional Ecological                                                                                                                                                                |

Information.

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. If peroxide formation is suspected, do not open or move container. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

**Storage.** Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place. Containers should be dated when opened and tested periodically for the presence of peroxides. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals. Keep away from heat, sparks and flame. Incompatible Materials. Strong bases. Oxidizing agent.

## 8. Exposure controls / personal protection

**Exposure Guidelines** This product does not contain any hazardous materials with occupational exposure limit established by the region specific regulatory bodies.

**Engineering Measures** Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

**Eye Protection**  
**Hand Protection**

Goggles  
Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Viton (R)      | recommendations   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly  
**Recommended Filter type:** Multi-purpose/ABEK conforming to EN14387 low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown

When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

No information available.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Liquid                   |
| Appearance                             | No information available |
| Odor                                   | No information available |
| Odor Threshold                         | No information available |
| pH                                     | No information available |
| Melting Point/Range                    | No data available        |
| Boiling Point/Range                    | No information available |
| Flash Point                            | No information available |
| Evaporation Rate                       | No information available |
| Flammability (solid,gas)               | Not applicable           |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | <=1100 hPa @ 50 °C       |
| Vapor Density                          | No information available |
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |
| Molecular Formula                      | C7 H6 Cl2 Mg             |
| Molecular Weight                       | 185.33                   |

### 10. Stability and reactivity

|                                  |                                                                                              |
|----------------------------------|----------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                   |
| Stability                        | Air sensitive. Moisture sensitive.                                                           |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition.                            |
| Incompatible Materials           | Strong bases, Oxidizing agent                                                                |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Hydrogen chloride, Magnesium oxides |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                     |
| Hazardous Reactions              | None under normal processing.                                                                |

### 11. Toxicological information

#### Acute Toxicity

#### Product Information

|                       |                                                                               |
|-----------------------|-------------------------------------------------------------------------------|
| Oral LD50             | Category 4. ATE = 300 - 2000 mg/kg.                                           |
| Dermal LD50           | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Vapor LC50            | Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.    |
| Component Information |                                                                               |

| Component               | LD50 Oral              | LD50 Dermal           | LC50 Inhalation      |
|-------------------------|------------------------|-----------------------|----------------------|
| 2-Methyltetrahydrofuran | 300-2000 mg/kg ( Rat ) | 4500 mg/kg ( Rabbit ) | 6000 ppm ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                        | CAS-No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------------------|----------|------------|------------|------------|------------|------------|
| 2-Methyltetrahydrofuran          | 96-47-9  | Not listed | Not listed | Not listed | Not listed | Not listed |
| 4-Chlorobenzylmagnesium chloride | 874-72-6 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

| Component               | Freshwater Algae                                | Freshwater Fish                                                  | Microtox   | Water Flea                                          |
|-------------------------|-------------------------------------------------|------------------------------------------------------------------|------------|-----------------------------------------------------|
| 2-Methyltetrahydrofuran | NOEC >= 104 mg/l (72h)<br>EC50 > 104 mg/l (72h) | LC50 (96h) > 100 mg/l<br>Onchorhynchus mykiss<br>(Rainbow trout) | Not listed | Chronic NOEC >=120 mg/l<br>(21 days, Daphnia magna) |

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

**DOT**

|                         |                                                           |
|-------------------------|-----------------------------------------------------------|
| UN-No                   | UN2924                                                    |
| Proper Shipping Name    | Flammable liquid, corrosive, n.o.s.                       |
| Technical Name          | (METHYLTETRAHYDROFURAN, 4-Chlorobenzylmagnesium chloride) |
| Hazard Class            | 3                                                         |
| Subsidiary Hazard Class | 8                                                         |
| Packing Group           | II                                                        |

**TDG**

|                         |                                     |
|-------------------------|-------------------------------------|
| UN-No                   | UN2924                              |
| Proper Shipping Name    | Flammable liquid, corrosive, n.o.s. |
| Hazard Class            | 3                                   |
| Subsidiary Hazard Class | 8                                   |
| Packing Group           | II                                  |

**IATA**

|                         |                                     |
|-------------------------|-------------------------------------|
| UN-No                   | UN2924                              |
| Proper Shipping Name    | Flammable liquid, corrosive, n.o.s. |
| Hazard Class            | 3                                   |
| Subsidiary Hazard Class | 8                                   |
| Packing Group           | II                                  |

**IMDG/IMO**

|                         |                                     |
|-------------------------|-------------------------------------|
| UN-No                   | UN2924                              |
| Proper Shipping Name    | Flammable liquid, corrosive, n.o.s. |
| Hazard Class            | 3                                   |
| Subsidiary Hazard Class | 8                                   |
| Packing Group           | II                                  |

## 15. Regulatory information

## International Inventories

| Component                        | CAS-No   | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|----------------------------------|----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| 2-Methyltetrahydrofuran          | 96-47-9  | X   | -    | X    | ACTIVE                                        | 202-507-4 | -      | -   |
| 4-Chlorobenzylmagnesium chloride | 874-72-6 | -   | -    | -    | -                                             | -         | -      | -   |

| Component                        | CAS-No   | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|----------------------------------|----------|-------|----------|------|------|------|------|-------|-------|
| 2-Methyltetrahydrofuran          | 96-47-9  | X     | KE-33479 | -    | X    | X    | X    | X     | X     |
| 4-Chlorobenzylmagnesium chloride | 874-72-6 | -     | -        | -    | -    | -    | -    | -     | -     |

**Legend:**

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component | Canada - National Pollutant | Canadian Environmental | Canada's Chemicals Management |
|-----------|-----------------------------|------------------------|-------------------------------|
|-----------|-----------------------------|------------------------|-------------------------------|

|                         | Release Inventory (NPRI) | Protection Agency (CEPA)<br>- List of Toxic Substances | Plan (CEPA) |
|-------------------------|--------------------------|--------------------------------------------------------|-------------|
| 2-Methyltetrahydrofuran | Part 4 Substance         |                                                        |             |

#### Other International Regulations

Authorisation/Restrictions according to EU REACH Not applicable

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                           | CAS-No   | OECD HPV       | Persistent Organic<br>Pollutant | Ozone Depletion<br>Potential | Restriction of<br>Hazardous<br>Substances (RoHS) |
|-------------------------------------|----------|----------------|---------------------------------|------------------------------|--------------------------------------------------|
| 2-Methyltetrahydrofuran             | 96-47-9  | Not applicable | Not applicable                  | Not applicable               | Not applicable                                   |
| 4-Chlorobenzylmagnesium<br>chloride | 874-72-6 | Not applicable | Not applicable                  | Not applicable               | Not applicable                                   |

| Component                           | CAS-No   | Seveso III Directive<br>(2012/18/EC) -<br>Qualifying Quantities<br>for Major Accident<br>Notification | Seveso III Directive<br>(2012/18/EC) -<br>Qualifying Quantities<br>for Safety Report<br>Requirements | Rotterdam<br>Convention (PIC) | Basel Convention<br>(Hazardous Waste) |
|-------------------------------------|----------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------|
| 2-Methyltetrahydrofuran             | 96-47-9  | Not applicable                                                                                        | Not applicable                                                                                       | Not applicable                | Not applicable                        |
| 4-Chlorobenzylmagnesium<br>chloride | 874-72-6 | Not applicable                                                                                        | Not applicable                                                                                       | Not applicable                | Not applicable                        |

## 16. Other information

**Prepared By** Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Revision Date** 01-April-2024  
**Print Date** 01-April-2024  
**Revision Summary** New emergency telephone response service provider.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**